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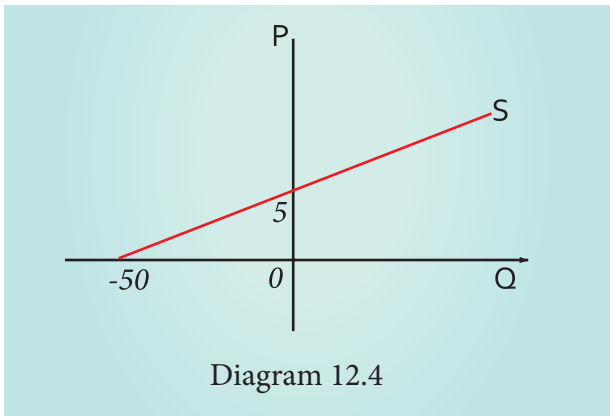


Diagram 12.4

Then $y-6=\frac{x-10}{10}$

$10(Y \bullet 6) \bullet X \bullet 10$

$10Y \bullet 60 \bullet X \bullet 10$

$10Y \bullet 60 \bullet 10 \bullet X$

$10Y \bullet 50 \bullet X$

$\bullet X \bullet \bullet 10Y \bullet 50$

Multiplying both sides by minus (•), we get

$X \bullet \bullet 50 \bullet 10Y$

Considering X as quantity supplied and Y as price (p)

Then $X \bullet 10p \bullet 50$ (or)

$$\boxed{X \bullet \bullet 50 \bullet 10\,p}$$

If Price = 0; Q = • 50

If Q = 0; P = 5

Note: The coefficient of 'p' is • in demand function.

The coefficient of 'p' is • in supply function.

12.2.4 Equilibrium

The point of intersection of demand line and supply line is known as equilibrium. The point of equilibrium is obtained by using the method of solving a set of equations. One can obtain the values of two unknowns with two equations. At equilibrium point,

Demand = Supply

(These are hypothetical examples)

$100 \bullet 10p \bullet 50 \bullet 10\,p$

$100 \bullet 50 \bullet 20p$

$50 \bullet 20p$

$\frac{50}{20} = p$

$p \bullet 2.5$

When $p \bullet 2.5$, Demand • $100 \bullet 10\,(2.5)$

$\bullet 75$

When $P \bullet 2.5$, Supply • $50 \bullet 10\,(2.5)$

$\bullet 75$

Example: 12.3

Find the equilibrium price and quantity by using the following demand and supply functions $Qd \bullet 100 \bullet 5P$ and $Q_s \bullet 5P$ respectively.

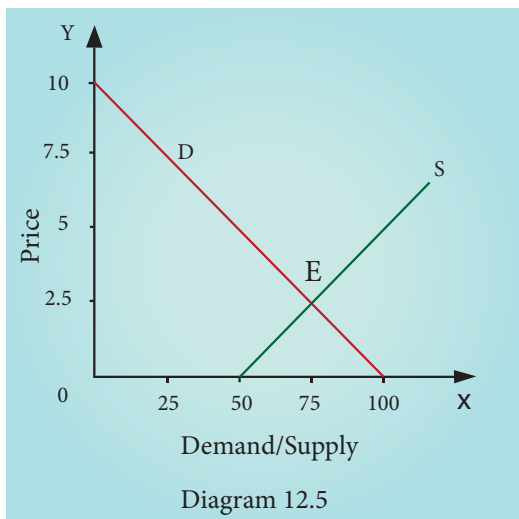


Diagram 12.5

Solution:

Equilibrium is attained when,

$Q_s \bullet Q_d$

$5P \bullet 100 \bullet 5P$

$10P \bullet 100$

$P \bullet 10$

When $P \bullet 10$

In supply function

$Q_s \bullet 5P \bullet 5 \times 10 \bullet 50$